

Miguel Ángel Arroyo

Multimedia Engineer

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Location: Alicante, Spain

PROFESSIONAL PROFILE

Multimedia Engineer with a strong foundation in web development, data engineering and AI integration. Hands-on experience designing data pipelines and applying large language models (LLMs) for information processing and structuring: RAG systems, agents, model fine-tuning and ETL processes. I combine technical and creative skills to bring value to innovative projects.

PROFESSIONAL EXPERIENCE

Internship — Data & AI Engineer · InferIA

July 2026 – Present

Development of data and AI pipelines for the collection, processing and structuring of information from public data sources.

- Designed and implemented data ingestion processes integrating diverse public data sources (business registries, patent and trademark offices, scientific repositories and public procurement portals) in heterogeneous formats (XML, JSON, CSV).
- Orchestrated the complete workflow with Apache Airflow deployed via Docker Compose on a GPU server: DAGs with task dependencies, scheduled biannual runs and model concurrency control through pools.
- Built web scrapers with requests and BeautifulSoup to extract the profiles of over 400 research groups from two universities along with their projects, respecting robots.txt and rate-limiting requests, with on-disk caching and retries with exponential backoff.
- Applied local large language models (LLMs) with Ollama/Llama 3.1 to filter, classify and extract structured information, normalizing it against a controlled catalog.
- Built robust data engineering processes: batch processing of large volumes, resumable caching, and idempotent data loads.
- Modeled and loaded data into PostgreSQL/PostGIS through version-controlled migrations with Alembic.
- Produced data source feasibility studies and technical documentation of the processes developed.

Technologies: Python, requests/BeautifulSoup, Ollama/Llama 3.1, Apache Airflow, Docker, PostgreSQL/PostGIS, Alembic, pandas, SQLite, Git.

SKILLS

Applied AI: RAG, agents, fine-tuning (LoRA/Dreambooth), ai-toolkit, technical prompt engineering, dataset creation, data labeling, vector databases, local LLMs (Ollama), AI-assisted development (Claude Code, Google Antigravity).

Data: ETL pipelines, orchestration with Apache Airflow, PostgreSQL/PostGIS, SQL, pandas, migrations (Alembic), batch processing, web scraping (requests/BeautifulSoup), API and public data source integration.

Web development: HTML, CSS, JavaScript, MEAN/MERN (MongoDB, Express, Angular/React, Node.js), Python, FastAPI, Git.

Game development: C++, Godot, Unreal Engine.

Design & art: Photoshop, Figma, Canva, Blender.

Soft skills: Agile methodologies, self-directed learning, aesthetic sense, technical problem-solving, technical communication.

FEATURED PROJECTS

Bachelor's Thesis — Generative Model Training Pipeline

Jan 2026 – Jul 2026

Development of a complete pipeline for training AI models focused on generating images under predefined artistic styles.

Technologies: Python, PyTorch, ai-toolkit, Diffusers, Fine-tuning (LoRA/Dreambooth).

m.ai — Personal Notes Assistant (WhatsApp Bot)

June 2026

Bot with a full RAG architecture that lets users save notes by text or voice, run semantic searches in natural language and generate intelligent summaries. Transcribes audio with Whisper, interprets images, auto-generates tags with an LLM, and uses a vector database with recency/relevance re-ranking.

Technologies: Python, FastAPI, Uvicorn, Groq API (gpt-oss-20b, Whisper, qwen3.6-27b), Upstash Vector, Meta Cloud API, Render.

RenderBox — Digital Assets Web Platform

Feb 2025 – May 2025

Web application designed to centralize the publishing, management and discovery of creative resources (2D, 3D, video and audio).

Technologies: MongoDB, Express, Angular, Node.js, Cloudinary, Multer, Vercel, Git.

EDUCATION

B.Eng. in Multimedia Engineering · University of Alicante

2022 – 2026

LANGUAGES

Spanish: Native. **English:** Fluent technical reading / Intermediate.